1)

Verstärkung 26 dB = $G_1$

BW bis 6 MHz (Signalbandbreite) = B

Rauschfaktor 1.6 dB = NF

$T = 290\,K$

Rauschfaktor in dB => $F = 10^{1.6/10} \approx 1.445$

$T_e = (F-1)\,T_0 = 290\,(1.445-1) \approx 129\,K$

$T_{ges} = T_0 + T_e = 289\,K$

$P_{out} = k\,B\,T_{out}\,G_1$

$= 9.54 \cdot 10^{-11}\,W$

in dB $\cdot 10 \lg\left(\frac{P}{1mW}\right)$

$\approx -70.2\,dBm$

b)

$R_{in} = 100\,\Omega$

$A = 22 \cdot 10^{-6}\,V$

$S_{in} = \frac{1}{2}\frac{v^2}{Z} = \frac{(22 \cdot 10^{-5})}{2 \cdot 100\,\Omega}$

$\approx -86.2\,dBm$

$S_{out} = G\,S_{in} \sim S_{out}$

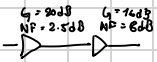
$= S_{in} + G = -50.2\,dBm$

$SNR = S_{out} - N_{out} = -50.2 - (-70.2)$

$= 20.0\,dB$

2)

$T_0 = 50\,K$ Rauschmaster




$B = 3\,MHz = 3 \cdot 10^3$

a) $T_{ges} = T_{01} + \frac{T_{02}}{G_1}$

$T_{e1} = 290\left(10^{\frac{NF}{10}} - 1\right) = 225.5\ K$

$T_{e2} = 290\left(10^{\frac{NF}{10}} - 1\right) = 1340\ K$

$10^{20/10}$

$T_{ges} = T_{e1} + \frac{T_{e2}}{G_1}$

$= 229.5 + \frac{1340}{100} = 244.1\ K$

b) essen nicht -90 dBm

$S_{out} = S_{in} + G_1 + G_2 = -54\ dBm$

$N_{out} = k(T_0 + T_{ges})\,B\,G \approx 4.8 \cdot 10^{-11}\ W$

$N_{out}[dB] = 10\log(4.8 \cdot 10^{-11}) \approx -73\ dB$

$SNR[dB] = S_{out} - N_{out} = 19.2\ dB$

c) mit kaskade

$SNR = S_{out} - N_{out}$

$= S_{in} + G_1 + G_2 + G_3 - 10\log\left(k(T_0 + T_{ges})B G\right)$

$= 17.2\ dB$